9 Building Complex Forms

Many apps grow beyond simple forms editing a single object. For example, when creating a `Person`you might want to allow the user to (on the same form) create multiple address records (home, work, etc.). When later editing that person the user should be able to add, remove or amend addresses as necessary.

9.1 Configuring the Model

Active Record provides model level support via the `accepts_nested_attributes_for` method:

```
class Person < ActiveRecord::Base
  has_many :addresses
  accepts_nested_attributes_for :addresses
end

class Address < ActiveRecord::Base
  belongs_to :person
end
```

This creates an `addresses_attributes=` method on `Person` that allows you to create, update and (optionally) destroy addresses.

9.2 Nested Forms

The following form allows a user to create a `Person` and its associated addresses.

```
<%= form_for @person do |f| %>
  Addresses:
  <ul>
    <%= f.fields_for :addresses do |addresses_form| %>
      <li>
        <%= addresses_form.label :kind %>
        <%= addresses_form.text_field :kind %>

        <%= addresses_form.label :street %>
        <%= addresses_form.text_field :street %>
        ...
      </li>
    <% end %>
  </ul>
<% end %>
```

When an association accepts nested attributes `fields_for` renders its block once for every element of the association. In particular, if a person has no addresses it renders nothing. A common pattern is for the controller to build one or more empty children so that at least one set of fields is shown to the user. The example below would result in 2 sets of address fields being rendered on the new person form.

```
def new
  @person = Person.new
  2.times { @person.addresses.build}
end
```

The `fields_for` yields a form builder. The parameters' name will be what `accepts_nested_attributes_for` expects. For example, when creating a user with 2 addresses, the submitted parameters would look like:

```
{
  'person' => {
    'name' => 'John Doe',
    'addresses_attributes' => {
      '0' => {
        'kind' => 'Home',
        'street' => '221b Baker Street'
      },
      '1' => {
        'kind' => 'Office',
```